

FINAL PROJECT DOCUMENTATION

PropTax – Online Property Tax Self-Assessment and Payment Portal

Software Engineering & Project Management

Student Name	Ragini Singh
Roll No.	150096724023
Subject	Software Engineering & Project Management
Case Study / PS No.	141
PS Title	PropTax – Online Property Tax Self-Assessment and Payment Portal
Project Type	Product Management + Software Development Case Study

Deployment Link

<https://proptax-property-tax-portal.vercel.app/>

GitHub Repository

[RaginiSingh2024 / PropTax Case Study Repository](#)

Figma / UI Design Link

<https://stitch.withgoogle.com/projects/12519224402147468096>

Current UI Prototype / Stitch

[PropTax UI Prototype](#)

Google doc link :

<https://docs.google.com/document/d/1g9Kua1PmE7Gc4bYEMrujKy8jndxS2j3po7Oj-PgJ3CA/edit?usp=sharing>

Final submission includes project planning, software engineering documentation, testing and risk management, together with a working frontend prototype.

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1. Project Overview

PropTax is a case-study based online property tax self-assessment and payment portal designed to improve the citizen journey from property search and tax assessment to online payment, receipt generation and objection handling. The project was developed as part of the Software Engineering & Project Management subject for Case Study No. 141.

The final project combines software requirements engineering, UML modelling, release planning, forecasting, software testing, risk management and a responsive HTML/CSS/JavaScript frontend prototype.

2. Problem Statement Context

The case study describes a city with a population of approximately 9 lakh and around 1.8 lakh properties. Property tax collection is mainly handled through ward offices. Although self-assessment is legally allowed, the existing process is paper-based and rate tables are difficult for property owners to understand, which can lead to disputes. The proposed portal provides online property search, self-assessment, tax calculation, online payment, receipts and an objection workflow.

3. Project Objectives

- Provide a clear online property search and self-assessment experience.
- Support transparent tax calculation through a dedicated calculator interface.
- Provide an online payment flow with payment confirmation and receipt access.
- Provide an objection submission and status-tracking interface.
- Document the software solution using SRS and UML artefacts.
- Plan delivery using backlog points, sprint velocity, milestones and release scope.
- Apply forecasting, test planning, defect analysis and risk management techniques.

4. Final Project Deliverables

Deliverable	Description
SRS Document	Requirements, scope, stakeholders, functional/non-functional requirements, MoSCoW and RTM.
UML Package	Use-case, class, sequence, activity and state diagrams with cohesion/coupling notes.
Project Plan	WBS, dependency/network view, critical dependency path, Gantt plan, sprints and milestones.
Forecast Sheet	Velocity-based release forecasting with and without the objections workflow.
Test Plan & Evidence	BVA, equivalence classes, decision table, test cases, defect metrics, DRE and RTM.
Risk Register & Closure	Risk identification, exposure, mitigation/RMMM actions, issue log and lessons learned.

Frontend Prototype

Responsive HTML5/CSS3/Vanilla JavaScript portal
covering the main citizen workflow.

5. Final Project Links

The following links are intended for final submission and demonstration. The GitHub repository and current UI prototype link are included below. The Vercel deployment URL and Figma URL should be pasted in the marked fields if they are different from the current project links.

Resource	Link
Deployment Link	https://proptax-property-tax-portal.vercel.app/
GitHub Repository	RaginiSingh2024 / PropTax Case Study Repository
Figma / UI Design Link	https://stitch.withgoogle.com/projects/12519224402147468096
Current UI Prototype / Stitch	PropTax UI Prototype

Submission note: The current UI prototype link available for the project is a Google Stitch project link. If the final design was exported or recreated in Figma, replace the Figma placeholder above with the actual Figma share URL.

6. Frontend Implementation

The frontend prototype was implemented using HTML5, CSS3 and Vanilla JavaScript. No backend or database is required for the prototype. Client-side demo behaviour uses JavaScript and localStorage where applicable. Sample tax values are presented as prototype/demo values because the case study does not provide an official municipal tax-rate table.

Main Frontend Screens / Modules

- Login / entry page and citizen dashboard
- Property search and property details
- Self-assessment form with validation
- Tax calculator with demo/prototype rate labelling
- Online payment simulation and payment-success flow
- Receipt and payment history
- Objection submission and status tracking
- Help and guidance pages

7. Frontend Screenshots

The following screenshots document the implemented PropTax frontend prototype. They are included as evidence of the user interface and major functional modules.

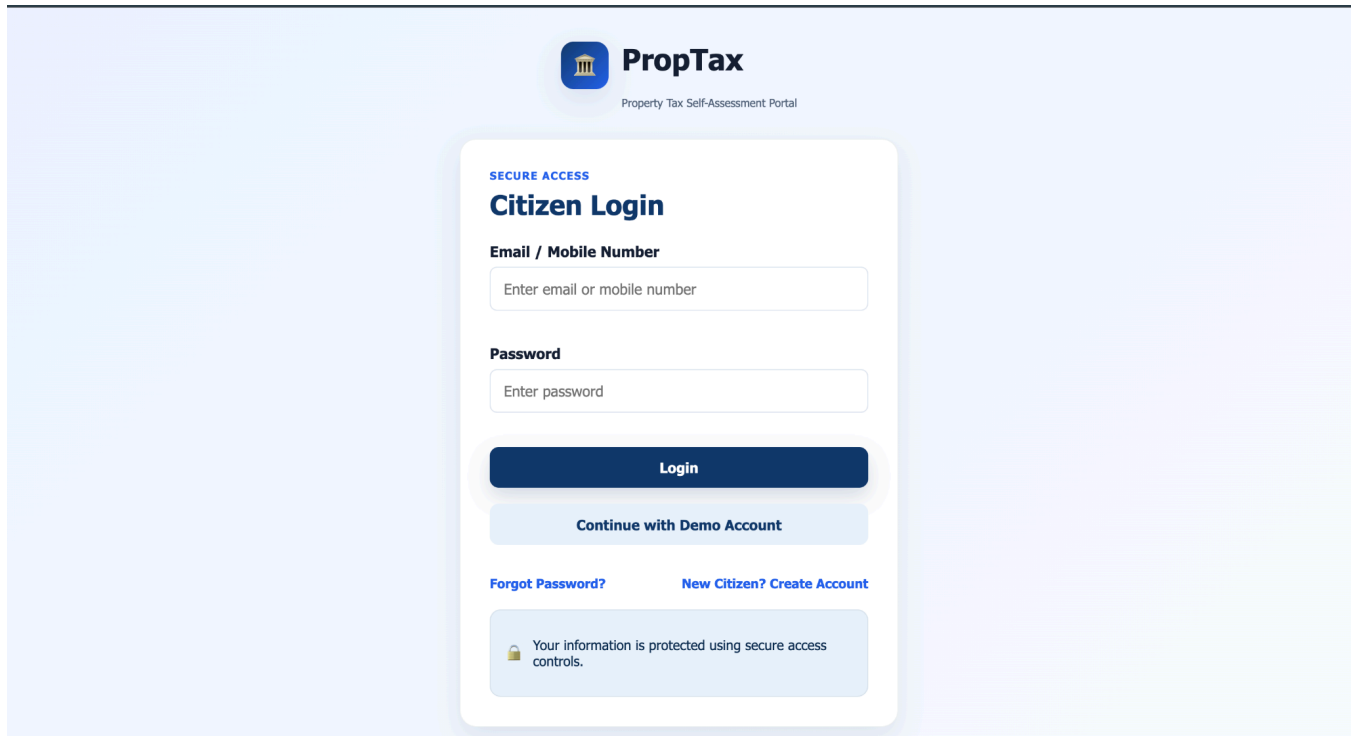


Figure 1. PropTax Objections page – objection submission and status workflow.

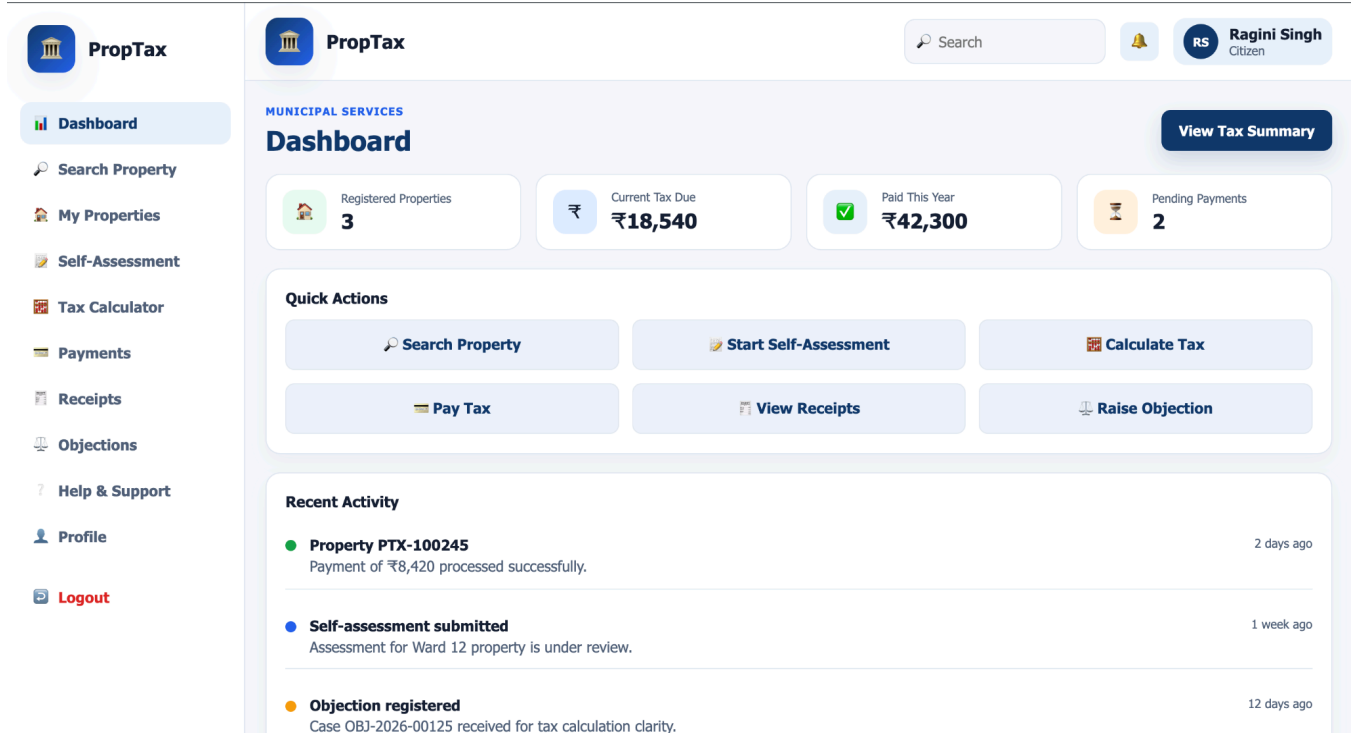


Figure 2. PropTax Tax Calculator – demo/prototype calculation interface.

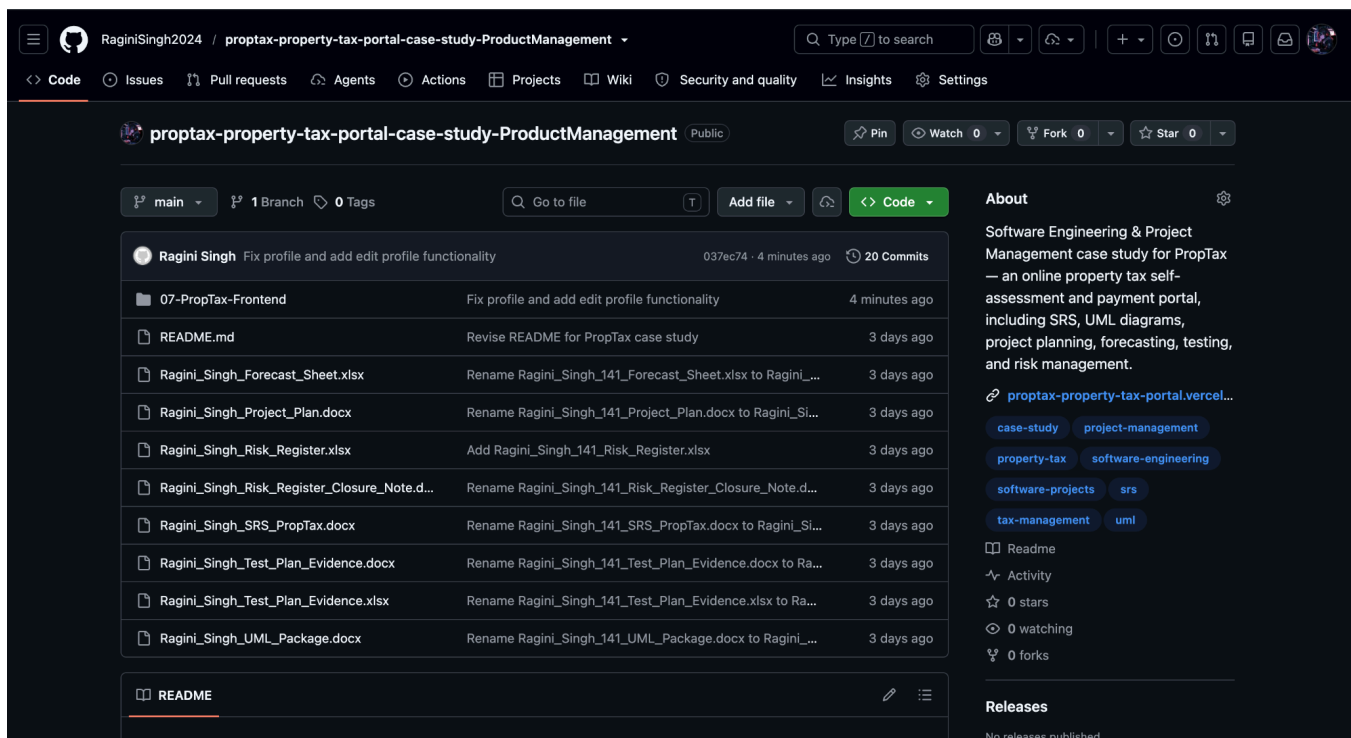


Figure 3. GitHub repository – project documentation and implementation files.



PropTax
Property Tax Self-Assessment Portal

SECURE ACCESS

Citizen Login

Email / Mobile Number

Enter email or mobile number

Password

Enter password

Login

Continue with Demo Account

Forgot Password?

New Citizen? Create Account



Your information is protected using secure access controls.



PropTax

Search

RS

Ragini Singh

Citizen

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

LEGAL REVIEW

Objections

Objection workflow may be subject to municipal/legal review.

My Objections

OBJ-2026-00125

Property: PTX-100245

Category: Tax Calculation

Subject: Adjustment mismatch in tax calculation

Status: Submitted

Submitted

Under Review

Decision Pending

Resolved

Raise New Objection

Property ID

PTX-100245

Objection Category

Select category

Subject

Brief description of objection

Description

Provide details of the issue

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PropTax

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

PropTax

Search



RS Ragini Singh
Citizen

PAYMENT PORTAL

Make Payment

Property Payment Summary

Property ID	PTX-100245
Owner	Ragini Singh
Tax Year	2026-27
Amount Payable	₹18,420

Choose Payment Method

☒ UPI

☐ Debit/Credit Card

☐ Net Banking

Pay Now

PropTax

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

PropTax

Search



RS Ragini Singh
Citizen

SUPPORT CENTER

Help & Support

How do I search for my property?

Go to Search Property and enter the property ID, owner name, ward, or address to view matching records.

How is self-assessment completed?

How can I calculate my property tax?

How can I pay my tax online?

Where can I download my receipt?

How do I raise an objection?

Contact Support

HELP DESK

EMAIL

PHONE

OFFICE HOURS

Mustang Pradesh Tax Support

support-demo@prontax.gov.in

+91 1800-123-4567

Mon - Sat 9:00 AM - 6:00 PM

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**PropTax**

Dashboard

Search Property

My Properties

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

**PropTax**

Search



RS

Ragini Singh
Citizen

ACCOUNT SETTINGS

Profile

Edit Profile

RS

Ragini Singh
Citizen

Active

Personal Information

First Name

Ragini

Last Name

Singh

Email Address

ragini@example.com

Mobile Number

9876543210

Address Information

Address

Demo Address

City

Mumbai

Ward

Ward 12

Account Information

**PropTax**

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

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Objections

Help & Support

Profile

Logout

**PropTax**

Search



RS

Ragini Singh
Citizen

PROPERTY RECORD

Property Details

Start Self-Assessment

PROPERTY SUMMARY

PTX-100245

Active

Owner Name

Ragini Singh

Address

12 MG Road, Sector 8

Ward

Ward 12

Property Type

Residential

Built-up Area

1200 sq.ft.

Property Age

8 years

Assessment Status

Updated

Current Tax Status

Pending

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**PropTax**

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

**PropTax**

Search



RS

Ragini Singh

Citizen

PAYMENT RECEIPT

Digital Receipt

Download Receipt

Print Receipt

**PropTax**

Dashboard

Search Property

My Properties

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

**PropTax**

Search



RS

Ragini Singh

Citizen

PROPERTY RECORDS

Search Property

Property ID

PTX-100245

Owner Name

Ragini Singh

Ward

Ward 12

Address

12 MG Road

Search

Reset

Matching Properties

PTX-100245

Owner: Ragini Singh

Address: 12 MG Road, Sector 8

Ward: Ward 12

Type: Residential

Status: Active

View Details

PTX-100312

Owner: Arjun Mehta

Address: 41 Ashok Vihar, Plot 9

Ward: Ward 7

Type: Residential

Status: Active

View Details

PTX-100480

Owner: Priya Nair

Address: 288 Market Street

Ward: Ward 4

Type: Commercial

Status: Active

View Details

PTX-100621

Owner: Vikram Sharma

Address: 14 Industrial Estate Road

Ward: Ward 18

Type: Industrial

Status: Active

View Details

PTX-100770

PTX-100905

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 PropTax

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

Objections

Help & Support

Profile

Logout

 PropTax

Search

RS Ragini Singh
Citizen

ASSESSMENT

Self-Assessment

1Property Details

2Classification

3Assessment Inputs

4Review

5Submit

Property ID

PTX-100245

Owner Name

Ragini Singh

Property Type

Residential

Built-up Area (sq.ft.)

1200

Property Age (years)

8

Ward

Ward 12

Usage Type

Self Occupied

Continue to Tax Calculation

 PropTax

Dashboard

Search Property

Property Details

Self-Assessment

Tax Calculator

Payments

Receipts

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Search

RS Ragini Singh
Citizen

SAMPLE CALCULATION

Tax Calculator

Property Type

Residential

Built-up Area (sq.ft.)

Property Age (years)

Usage Type

Self Occupied

Calculate

Demo / Prototype Rate • Sample calculation for demonstration

Review & Pay

8. Project Planning & Forecast Summary

The case study provides a backlog of 34 points for Search & Calculator, 21 points for Payment & Receipts, and 29 points for the Objections Workflow. The given sprint velocity is 11 points per two-week sprint.

Scope	Points	Sprints Required	Duration
Search & Calculator + Payment & Receipts	55	5	10 weeks
Full scope including Objections	84	8	16 weeks

The Release 1 planning therefore keeps the full objections workflow outside the initial release, leaving a six-week schedule buffer under the 16-week tax-year constraint. The objections workflow can be planned as a subsequent release or included only if scope and legal requirements are confirmed.

9. Testing & Quality Approach

- Boundary Value Analysis for relevant input boundaries.
- Equivalence Class Partitioning for valid and invalid input categories.
- Decision-table based testing for payment and workflow conditions.
- Functional tests for search, self-assessment, calculation, payment, receipts and objections.
- Security-oriented checks for authentication and unauthorized access.
- Defect logging and worked-example quality metrics.

10. Risk Management Summary

- Legal objection requirements may change or remain unclear.
- Tax rate and rebate rules must be validated against approved municipal rules.
- Payment gateway integration and confirmation must be tested early.
- Lower-than-planned velocity can affect release timing.
- Property data quality and security require controls and validation.

11. Project Outcome

The final PropTax submission demonstrates an end-to-end software engineering and project management approach. It connects requirements engineering with UML design, agile release planning, forecasting, software testing, risk management and a working frontend prototype. The project also provides a demonstrable UI that can be accessed through the deployment and design links listed in Section 5.

12. Final Submission Checklist

- ☐ Student details included
- ☐ Subject and Case Study / PS details included
- ☐ Project title documented
- ☐ SRS completed
- ☐ UML package completed
- ☐ Project plan and sprint forecast completed
- ☐ Forecast sheet completed
- ☐ Test plan and evidence completed
- ☐ Risk register and closure note completed
- ☐ Frontend prototype completed
- ☐ GitHub repository linked
- ☐ Deployment link to be inserted
- ☐ Figma link to be inserted
- ☐ Frontend screenshots included

— End of Final Project Documentation —